

Computer Vision & Cognitive Systems

SCQ5109806 - LM CS,DS,CYB,PD



Local features: visual (logo) recognition contest

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SIFT matching

- An example:



Visual Recognition Contest

- **Task:** instance-based object recognition
 - ▶ Test different combinations of local visual features and image matching strategies
 - ▶ Participate by submitting your solutions and check the leaderboard/results (links on Moodle)



- ▶ **Query images:** 4 logos
- ▶ **Dataset** (testset): 110 images
- ▶ **Evaluation metrics:** precision, recall, mean avg precision (mAP)

Visual Recognition Contest

- ▶ Deadline: November 11, 2025 - 11:00pm
- ▶ You can participate individually or as a group (2 persons)
- ▶ In your entries (max 3 submissions) as username put full name(s) and add a short description of what you have done in the note field
- ▶ Top-ranked groups shall present their approach (~2 min)



- ▶ **Query images:** 4 logos
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Evaluation metrics

- **Precision** and **Recall** are the standard metrics used for evaluating information retrieval systems
 - $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$
 - $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$
- An example:

		+		-	
		In Result		Not in result	
+	Relevant	30 (TP)		20 (FN)	
-	Not Relevant	10 (FP)		40 (TN)	

Groundtruth

Output

$$\text{Prec} = 30 / (30 + 10) = 0.75$$

$$\text{Rec} = 30 / (30 + 20) = 0.60$$

Evaluation metrics

- It is common to plot *Precision-Recall curves* (or ROC) for different cut-off points of a parameter
- For systems that return a ranked list *Average Precision (AP)* is the most common evaluation metric
 - We can plot precision as a function of recall r : i.e. $p(r)$
 - AP is the average value of $p(r)$ over the interval from $r=0$ to $r=1$ (i.e. it computes the area under the curve)

$$\text{AveP} = \frac{\sum_{k=1}^n (P(k) \times \text{rel}(k))}{\text{number of relevant documents}}$$

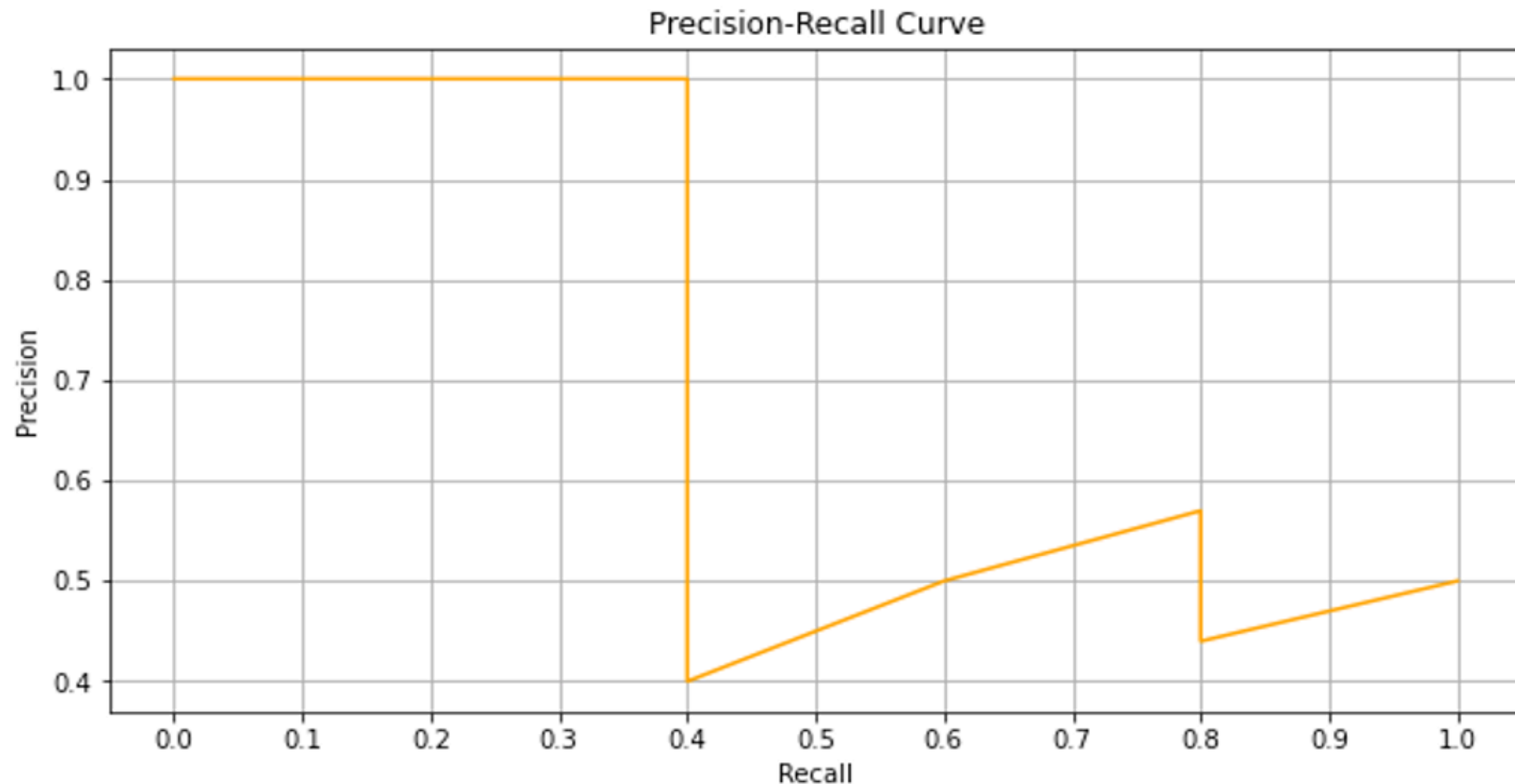
Evaluation metrics

- AP, an example:
 - ▶ Let's consider a dataset of 10 images (5 containing the searched logo, e.g. *Coca Cola*):

Rank	Score	Correct?	Precision	Recall
1	210	TRUE	1.0	0.2
2	150	TRUE	1.0	0.4
3	147	FALSE	0.67	0.4
4	135	FALSE	0.5	0.4
5	122	FALSE	0.4	0.4
6	80	TRUE	0.5	0.6
7	75	TRUE	0.57	0.8
8	45	FALSE	0.5	0.8
9	23	FALSE	0.44	0.8
10	12	TRUE	0.5	1.0

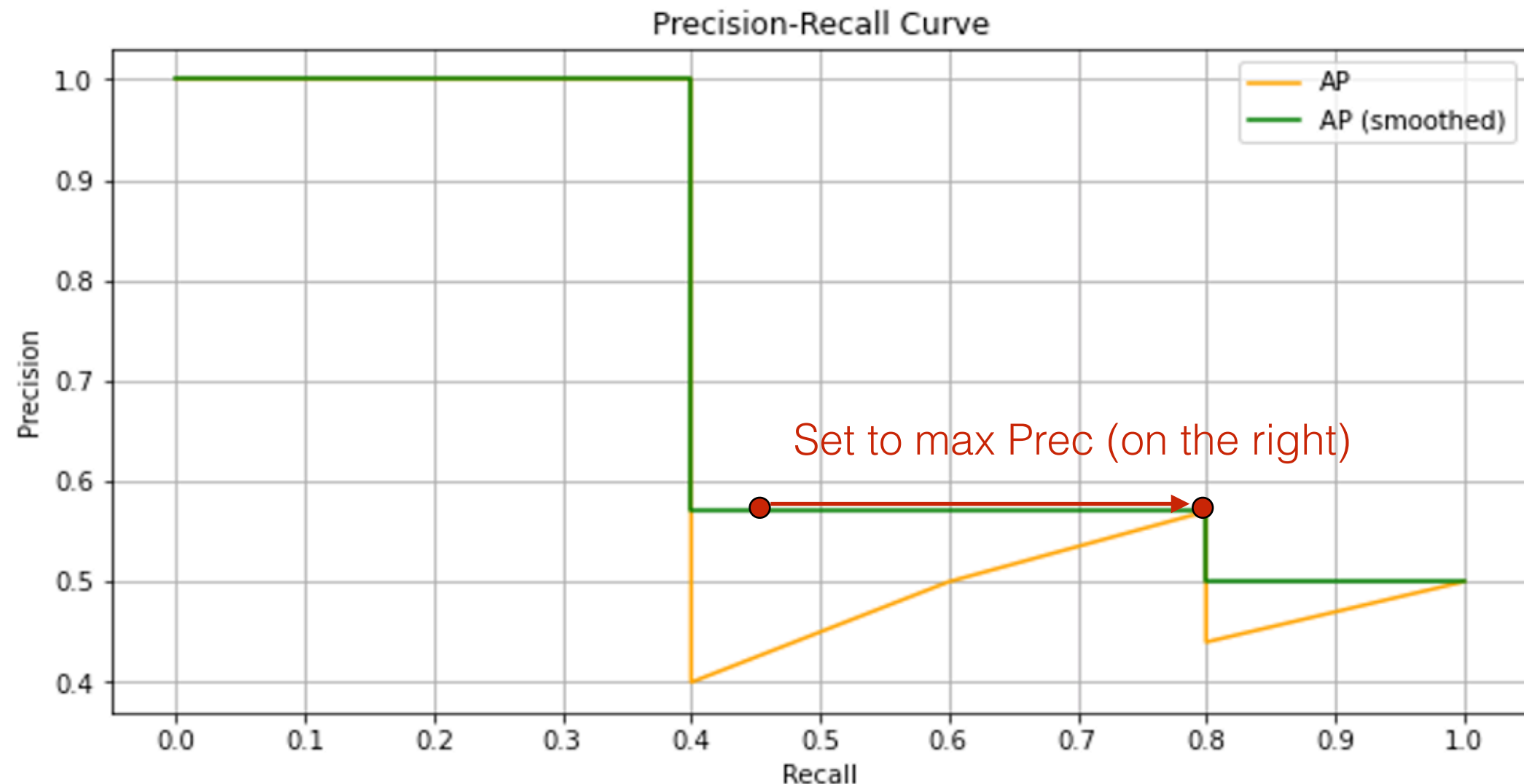
Evaluation metrics

- From AP to Interpolated AP (through an example):



Evaluation metrics

- From AP to Interpolated AP (through an example):

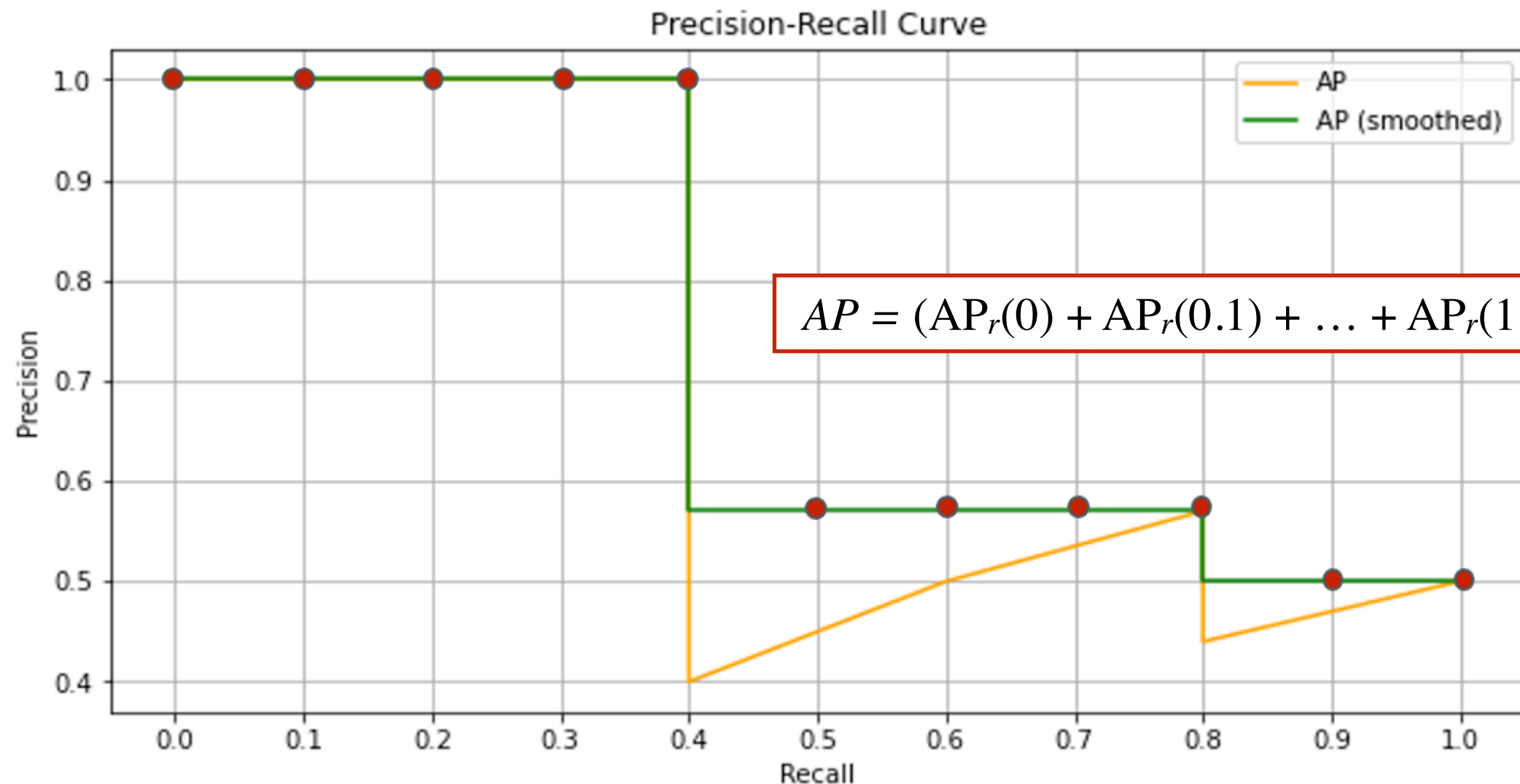


Evaluation metrics

- From AP to Interpolated AP (through an example):

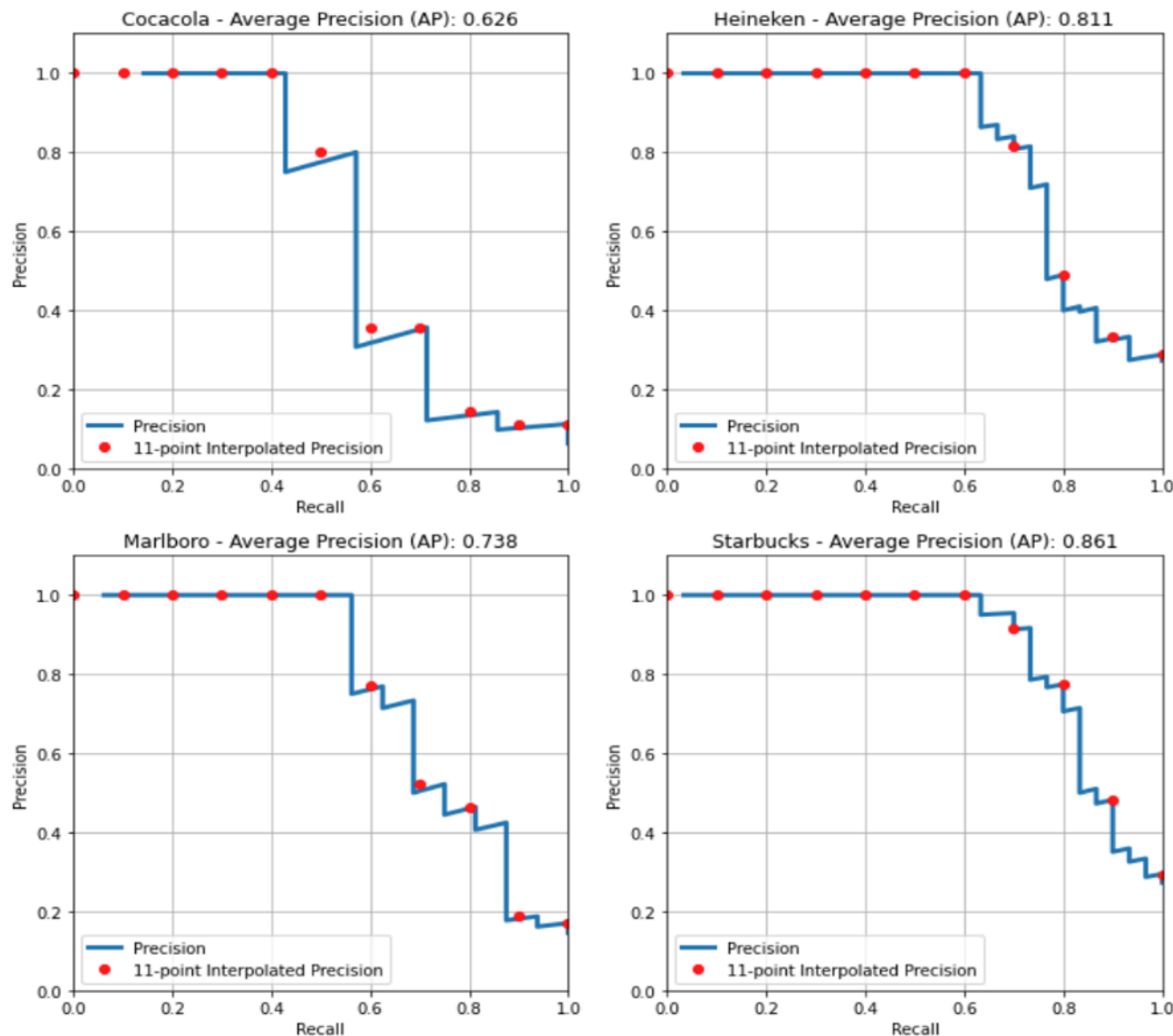
Discretize the recall values from 0 to 1.0 into 11 points: 0, 0.1, 0.2,

Next, compute the average of ma precision value for these recall values



Visual Recognition Contest

- Our baseline:



Our pipeline:

- Standard SIFT (DoG, 128-d descriptors)
- SIFT Matching
- Threshold on #matches

(Interpolated AP)

Average Precision (AP):

Cocacola	==>	0.626
Heineken	==>	0.811
Marlboro	==>	0.738
Starbucks	==>	0.861

Mean Average Precision (mAP) ==> 0.759

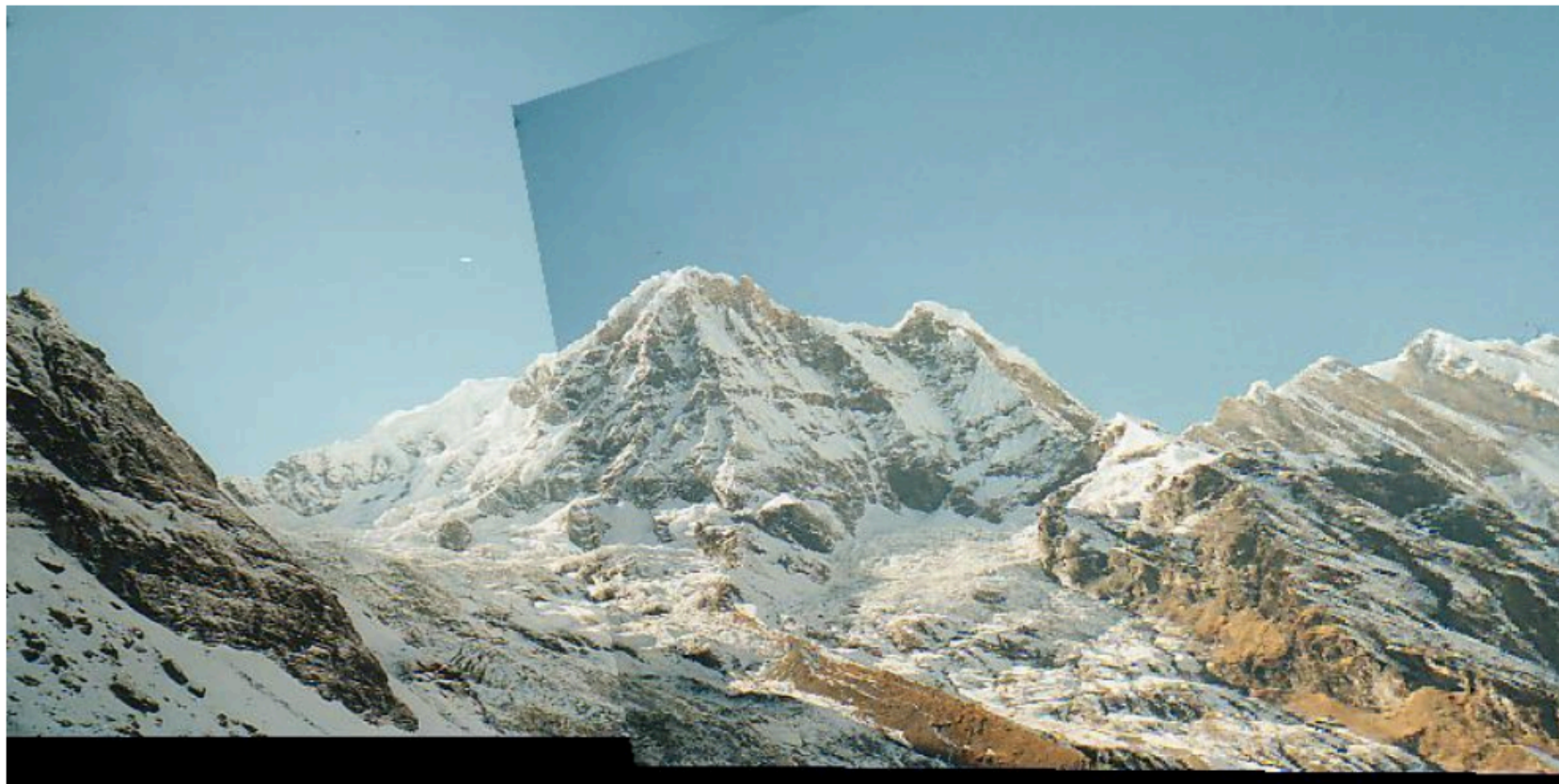
Visual Recognition Contest

- A few advice:
 - Check and try different parameters of the SIFT algorithm (both detection, e.g. octaves, and description)
 - Try different local visual features (e.g. SURF, ORB, ...) or different combinations of detectors + descriptors

Feature Detector	Corner	Blob	Region	Rotation invariant	Scale invariant	Affine invariant	Repeatability	Localization accuracy	Robustness	Efficiency
Harris	✓			✓			+++	+++	+++	++
Hessian		✓		✓			++	++	++	+
SUSAN	✓			✓			++	++	++	+++
Harris-Laplace	✓	(✓)		✓	✓		+++	+++	++	+
Hessian-Laplace	(✓)	✓		✓	✓		+++	+++	+++	+
DoG	(✓)	✓		✓	✓		++	++	++	++
SURF	(✓)	✓		✓	✓		++	++	++	+++
Harris-Affine	✓	(✓)		✓	✓	✓	+++	+++	++	++
Hessian-Affine	(✓)	✓		✓	✓	✓	+++	+++	+++	++
Salient Regions	(✓)	✓		✓	✓	(✓)	+	+	++	+
Edge-based	✓			✓	✓	✓	+++	+++	+	+
MSER			✓	✓	✓	✓	+++	+++	++	+++
Intensity-based			✓	✓	✓	✓	++	++	++	++
Superpixels			✓	✓	(✓)	(✓)	+	+	+	+

Visual Recognition Contest

- A few advice:
 - Try different features and “classification” strategies...



Step 1: detect
keypoints

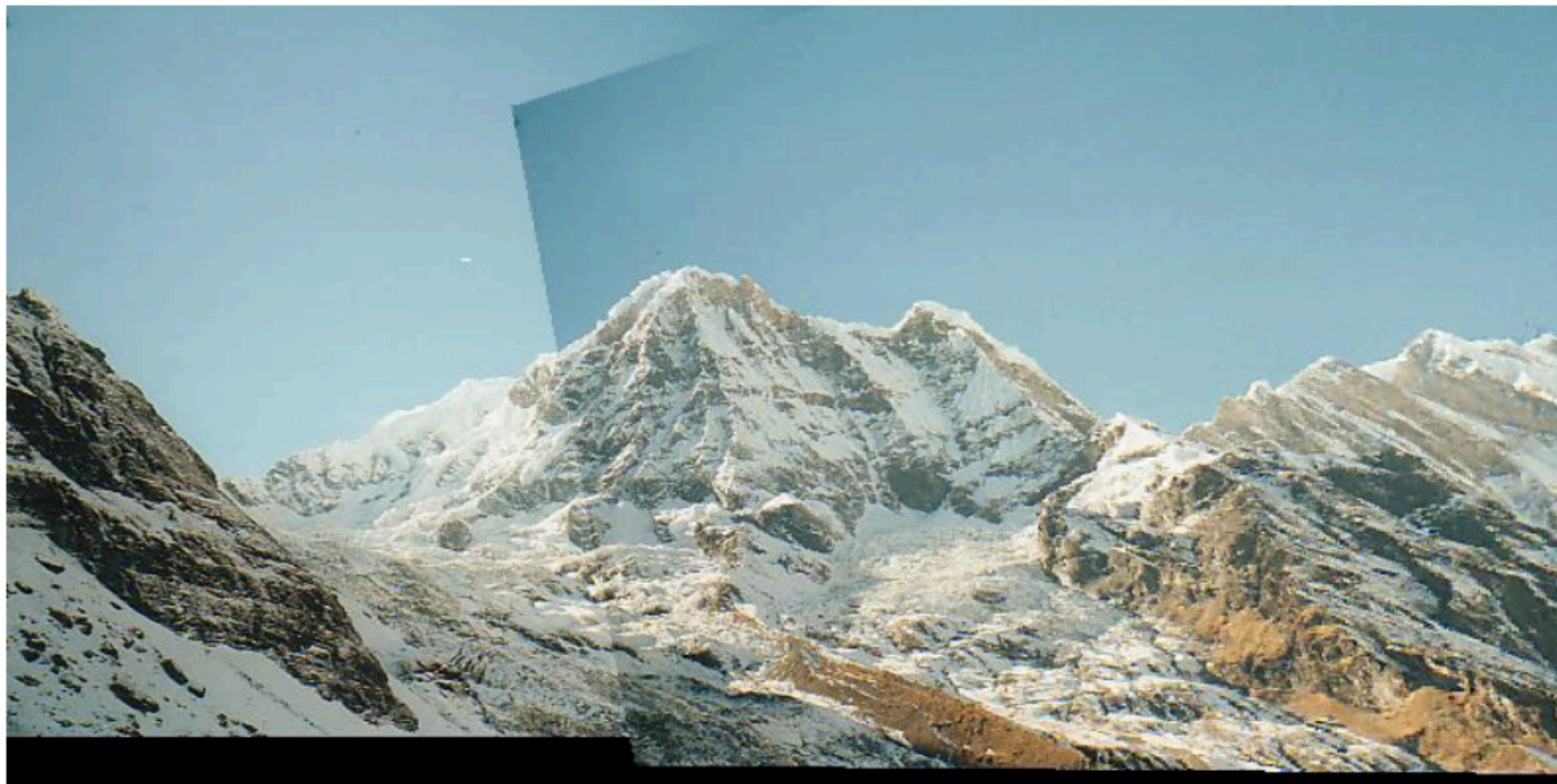
Step 2: build
keypoints descriptors

Step 3: match
keypoint features

Step 4: align images (i.e.
fitting the transformation)

Visual Recognition Contest

- A few advice:
 - Try different features and “classification” strategies... as well as better image matching approaches



Step 1: detect keypoints

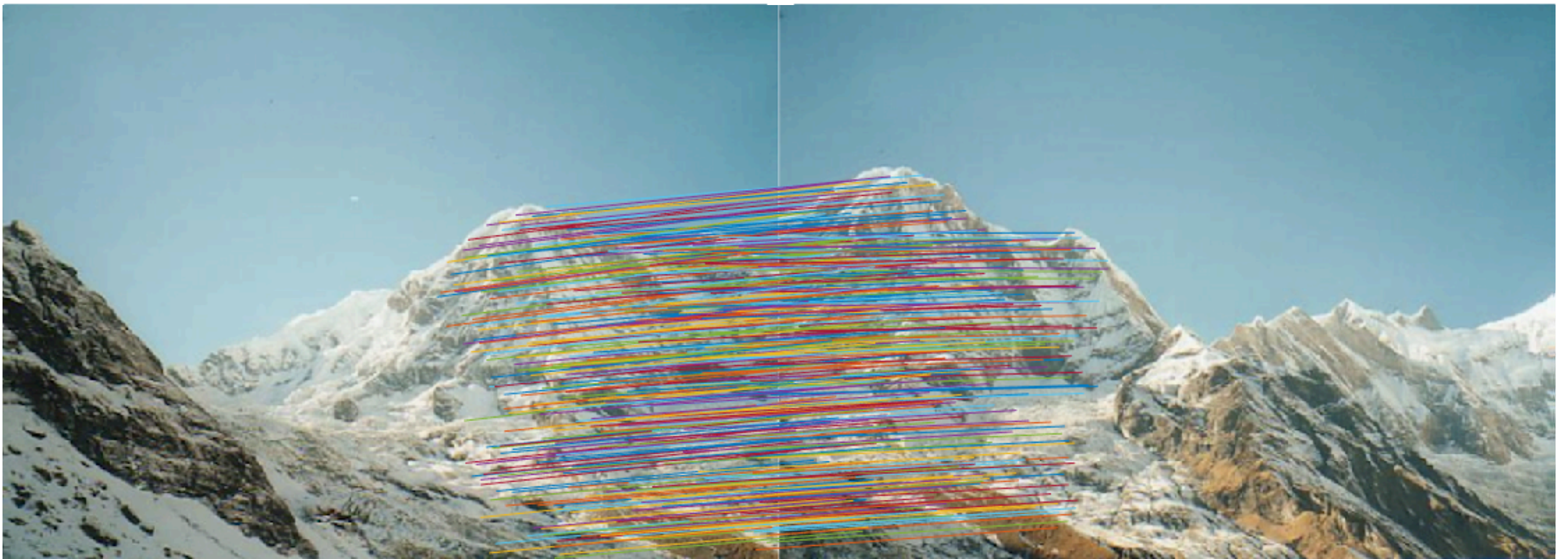
Step 2: build keypoints descriptors

Step 3: match keypoint features

Step 4: align images (i.e. fitting the transformation)

Geometric verification

- A bit more on “step 4”, i.e. *geometric verification*
 - The idea is to test whether matches are consistent with an overall image transformation
 - Inconsistent matches are rejected



Contact

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